

SHASHI BHUSHAN JHA

Wireless Communications Researcher

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RESEARCH PROFILE

M.Tech graduate in Electrical Engineering with a specialization in Communication and Signal Processing. My research focuses on analytical modeling and simulation of wireless multiple-access systems, PHY-layer receiver design, bit-error-rate analysis, and successive interference cancellation. I am seeking research and doctoral opportunities in wireless communications and emerging 6G systems.

RESEARCH INTERESTS

- Wireless communications and signal processing
- 6G and beyond
- Non-terrestrial networks (NTN)
- Integrated sensing and communications (ISAC)
- Multiple access and NOMA
- PHY-layer receiver algorithms

EDUCATION

Master of Technology in Electrical Engineering

2024–2026

Indian Institute of Technology Ropar

CGPA: 7.78/10

Specialization: Communication and Signal Processing

Relevant Coursework: RF Systems for Communication; Analytical Techniques for Signal Processing; Detection and Estimation Theory

Thesis: "Generalization of Downlink and Uplink NOMA Framework for Future Generation Technology"

Supervisor: Dr. Satyam Agarwal

Bachelor of Engineering in Electrical Engineering

2016–2022

Purbanchal University, Biratnagar, Nepal

CGPA: 2.98/4.00

Major: Power Systems

MANUSCRIPT IN PREPARATION

Analytical BER Study with Hardware Validation for N-User NOMA

Internal review

Prepared for submission to an IEEE Transactions journal; currently undergoing internal faculty review.

RESEARCH EXPERIENCE

Master's Thesis Research — Power-Domain NOMA

Jul 2025–May 2026

Indian Institute of Technology Ropar

Communication and Signal Processing

- Developed a unified BER and throughput framework for 2–4-user uplink and downlink power-domain NOMA over flat Rayleigh fading with explicit SIC-error modeling.
- Derived closed-form two-user downlink BER expressions and SIC-consistent uplink and generalized multi-user formulations.
- Evaluated power allocation, distance-based user ordering, SIC depth, NOMA–OMA throughput, and oversampling through Monte Carlo simulation.
- Showed how oversampling reduces effective noise variance while user ordering and error propagation continue to govern SIC reliability.

Research Intern — Autonomous Systems

Sep 2025–Mar 2026

SkyFlock Uavation Pvt. Ltd.

Remote

- Studied swarm-drone architectures and inter-drone communication protocols.
- Used ROS 2, Gazebo, PX4, and MAVSDK for multi-agent simulation, control, and telemetry analysis.

PROFESSIONAL EXPERIENCE

- Trainee Associate — Prosecution and Drafting Department**
InventIP Legal Services LLP

18 May 2026–12 Aug 2026
Gurugram
- Worked in the Prosecution and Drafting Department, gaining professional experience with patent prosecution and drafting workflows.
- Instructor — Computer Science and Mathematics**
Delhi Public School, Dharan

Apr 2022–Dec 2023
Nepal
- Taught Python programming, computational thinking, and mathematics; developed course materials and mentored senior students.

SELECTED TECHNICAL PROJECTS

- Rat-Race Hybrid Coupler**

Course project
- Designed and simulated a rat-race hybrid coupler in Ansys HFSS for the RF Systems for Communication course.
- Echo Cancellation Using Adaptive Filtering**

Jan–May 2025
- Implemented normalized least-mean-square adaptive filtering in MATLAB to suppress acoustic echo and assess convergence and signal-quality behavior.
- Audio Signal Analysis**
FFT, spectrogram, and MFCC-based time-frequency analysis using Python, SciPy, and Librosa.

2024

Smart Surveillance
Real-time motion detection and object tracking using Python, OpenCV, edge detection, and frame differencing.

2024

TECHNICAL SKILLS

- Wireless and DSP**

Power-domain NOMA, SIC, BER analysis, Rayleigh fading, communication theory, detection and estimation, adaptive filtering
- Research Methods**

Analytical modeling, Monte Carlo simulation, performance characterization, technical writing, literature review
- Programming**

MATLAB, Python, C ++; NumPy, Pandas, SciPy, scikit-learn, OpenCV
- Tools and Platforms**

Ansys HFSS, Linux, Git/GitHub, ROS 2, Gazebo, PX4, MAVSDK, Docker

ADDITIONAL INFORMATION

- Certifications**

WIPO General Course on Intellectual Property; WIPO Essentials of Patents; CS105: Introduction to Python; Introduction to the Internet of Things
- Languages**

English (professional), Hindi (full professional), Nepali (native/bilingual), Maithili (native/bilingual)